

脉冲神经网络（SNN）训练中的信息丢失与恢复： 中文汇总与研究思路

Tao

摘要

I. 文章汇总

E-prop: A solution to the learning dilemma for recurrent networks of spiking neurons [1];

OTTT: Online Training Through Time for Spiking Neural Networks [2];

NDOT: NDOT: Neuronal Dynamics-based Online Training for Spiking Neural Networks [3];

S-TLLR: S-TLLR: STDP-inspired Temporal Local Learning Rule for Spiking Neural Networks [4];

TESS: TESS: A Scalable Temporally and Spatially Local Learning Rule for Spiking Neural Networks [5]

II.

参考文献

- [1] G. Bellec, F. Scherr, A. Subramoney, E. Hajek, D. Salaj, R. Legenstein, and W. Maass, "A solution to the learning dilemma for recurrent networks of spiking neurons," *Nature Communications*, vol. 11, no. 1, p. 3625, Jul. 2020. [Online]. Available: <https://doi.org/10.1038/s41467-020-17236-y>
- [2] M. Xiao, Q. Meng, Z. Zhang, D. He, and Z. Lin, "Online Training Through Time for Spiking Neural Networks," in *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., vol. 35. Curran Associates, Inc., 2022, pp. 20717–20730. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/file/82846e19e6d42ebfd4ace4361def29ae-Paper-Conference.pdf
- [3] H. Jiang, G. D. Masi, H. Xiong, and B. Gu, "NDOT: Neuronal Dynamics-based Online Training for Spiking Neural..." [Online]. Available: <https://openreview.net/forum?id=eIF0QoBSFV>
- [4] M. P. E. Apolinario and K. Roy, "S-TLLR: STDP-inspired Temporal Local Learning Rule for Spiking Neural Networks," Oct. 2024, arXiv:2306.15220 [cs]. [Online]. Available: <http://arxiv.org/abs/2306.15220>
- [5] M. P. E. Apolinario, K. Roy, and C. Frenkel, "TESS: A Scalable Temporally and Spatially Local Learning Rule for Spiking Neural Networks," in *2025 International Joint Conference on Neural Networks (IJCNN)*, Jun. 2025, pp. 1–9, iSSN: 2161-4407. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/11227652>